

SEAN PREVITY

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EDUCATION

University of Central Florida, Computer Science, M.S. 2026 — Present

University of Central Florida, Computer Science, B.S. 2021 — 2025

- Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence Fundamentals, Computer Architecture, Software Engineering, Discrete Mathematics
- GPA: 3.64

PROJECTS

European Football Stats October 2025 - February 2026

- Created a full-stack analytics platform for the top 5 European football leagues, supporting match previews, historical analysis, and predictive insights.
- Built with **Next.js**, **Node.js**, and **PostgreSQL (Supabase)** with seamless data updates.
- Engineered a **Monte Carlo simulation** model using contextual match and team data to generate probabilistic match outcome predictions.
- Designed a relational data model supporting **200+** teams, **3,000+** players, **35,000+** matches, and **20+** years of historical league data.

France Travel Guide April — September 2025

- Designed and deployed a **bilingual** (French/English) travel site using Next.js, Node.js, and PostgreSQL with **37,000** rows of public INSEE data.
- Integrated multiple third-party APIs to aggregate and present dynamic city data and media content.
- Reduced API costs and latency by **45%** by implementing an AI description caching system.
- Deployed a hybrid **AWS** architecture combining serverless APIs (**API Gateway**), relational storage (**RDS**), and **EC2**-based services, with **Amplify** and **Cognito** for frontend hosting and authentication.

Pegasus Parking October - December 2024

- Developed a MERN application for coordinating university parking availability.
- Designed **RESTful** API endpoints to support concurrent users during peak campus traffic periods.
- Built a modular **Express** backend with **Mongoose** for scalable data modeling and API endpoints.
- Integrated Google Maps Routes API to compute real-time directions and dynamically select the closest parking location from 14 campus sites.

Babel January — July 2025

- Developed a proof-of-concept translation platform with a React frontend and FastAPI backend.
- Collaborated in an **Agile** team using **Jira sprints**.
- Researched and benchmarked open-source translation models using HuggingFace with COMET and BLEU evaluation metrics.
- Fine-tuned translation models, achieving up to a **35%** accuracy improvement on multiple models.
- Prompt engineered multiple AI translation models to generate structured, consistent outputs, enabling a reliable automated benchmarking pipeline.
- Created document parsing and layout-preservation logic for DOCX and PDF files.

SKILLS

- **Programming Languages:** Java, Python, JavaScript, TypeScript, C/C++
- **Frameworks:** React, Next.js, Node.js, Express.js, Redux, Flask, FastAPI
- **Databases/Cloud:** PostgreSQL, MySQL, MongoDB, Supabase, AWS, Docker, Drizzle ORM
- **Tools:** Git, GitHub, Postman, Jira, Teams
- **Data Structures & Algorithms:** Top 15% in contests on LeetCode with **600+** Problems solved
- **Languages:** English (Native), French (Limited Working)